Modeling of Seasonal Dynamics for Traffic-Dominant Particulate Matter Pollution in The Indo-Gangetic Plain: Using Multi-Scale Wavelet Analysis, Source Apportionment, And Episode-Specific Control Strategies

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Abstract: In terms of PM_2, the Indo-Gangetic Plain in India fluctuates greatly. During the winter, pollution levels are 140–180 μg/m³, whereas during the monsoon, they drop to 35–55 μg/m³. The regulations for reducing air pollution hardly alter with the seasons, despite the 4.5–5 times difference. We investigated Delhi's air over a five-year period (2019–2024) using a variety of techniques, including rolling Positive Matrix Factorisation, k-means clustering, continuous wavelet analysis, and some precise, season-focused machine learning. The concept? Determine the best time and way to take action. Three primary rhythms were identified by the wavelet analysis: a large one associated with the seasons (about 200 days), a mid-sized one during the transitions (75–100 days), and a smaller cycle that monitors the weather (10–40 days). Five distinct categories of pollution episodes, each with a unique combination of sources and weather patterns, were identified by k-means clustering. Rolling PMF revealed something startling: because all that stagnant air absorbs pollution, traffic's percentage of $PM_2.^1$ increases dramatically in the winter, reaching 35–50%. Even though the annual average is 22 ± 8%, traffic's proportion falls to 10–15% during the monsoon due to improved ventilation. The Shannon entropy indicates the degree of source dispersion. Targeted restrictions are effective since only two or three sources account for 65–80% of the pollution in winter (entropy of 1.6–1.9 bits). The monsoon changes everything; sources become more dispersed (entropy of 2.7–3.1 bits), necessitating the use of more comprehensive and integrated techniques. People tend to forget that 70–85% of PM caused by vehicles comes from non-exhaust sources such road dust, tyre wear and brake dust. The apparent decline in $PM_2.^2$ (−2.8 μg/m³ annually, $p<0.05$) disappeared when we adjusted for weather (deweathering). The actual, weather-adjusted trend is not statistically significant and is significantly weaker (−1.2 ± 0.8 μg/m³ year, $p>0.05$). Generic, year-round machine learning models were far less effective than season-specific ones. In comparison to a year-round R^2 of 0.74, accuracy increased by 20–25% (winter $R^2=0.89$, monsoon $R^2=0.93$). What does this imply for policy, then? PM can be reduced by 30–40% with targeted winter interventions such limiting heavy trucks, regulating heating, and reducing ammonia. There is a two-month window immediately following the monsoon when reducing biomass burning can reduce sources by 50–70%. By combining these tactics, annual PM can be reduced by 40–50%, bringing Delhi closer to the WHO's interim objective of 35 μg/m³ by 2028–2030. This study provides a strong, data-driven road map for more

intelligent, season-aware air quality management, shifting from broad, year-round controls to more flexible, successful interventions.

Keywords: *Traffic emissions, source apportionment, particulate matter (pm$_{2.5}$), seasonal variation, wavelet analysis, shannon entropy*

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1. Introduction

With winter PM$_{2}$.^{1} concentrations (140–180 μg/m^3) 4.5–5 times higher than monsoon levels (35–55 μg/m^3), the Indo-Gangetic Plain (IGP) displays extraordinary seasonal particulate matter fluctuation unmatched in magnitude among worldwide urban locations (Chauhan et al., 2022; Ubreathe Air Purifiers, 2024). Compared to temperate metropolitan areas, this extraordinary seasonality and a rapidly motorising vehicle fleet—which is predicted to reach 500 million vehicles by 2050—require a fundamentally different understanding of atmospheric chemistry and meteorology (Nature Scientific Data, 2025). According to the Health Effects Institute (2025) and Harvard School of Public Health (2024), long-term exposure to air pollution in India increases mortality by 1.5 million deaths per year when compared to WHO-recommended levels. In 2023, air pollution contributed 7.9 million deaths worldwide as the second leading mortality risk factor. The connection between dose and response is especially strong: every 10 μg/m^3 increase in PM$_{2}$.^{1} is associated with an 8.6% increase in all-cause mortality (Jaganathan et al., 2024). According to current NAAQS standards (2009–2019), PM$_{2}$.^{1} is responsible for about 3.8 million deaths, which rises to 16.6 million deaths under WHO guidelines (5 μg/m^3). With IGP concentrations reaching 119 μg/m^3 (23 times the WHO recommended limit), the whole Indian population suffers from year-round air pollution that exceeds WHO guidelines.

In India's air quality crisis, road traffic is an important but little-studied seasonal factor. 20–30% of urban ambient PM$_{2}$.^{1} worldwide is caused by vehicle emissions; real-time mapping shows significant spatial variability, with emission intensity ranging by factors of up to three across 1 km road segments (Wang et al., 2025). According to Bharat Stage VI norms, heavy commercial vehicles generate nitrogen oxides 8.5 ± 2.1 times more than light vehicles, with 70–85% of traffic-related PM$_{2}$.^{1} coming from non-exhaust particulate matter (brake wear, tire abrasion, road resuspension) (Wang et al., 2025). Respiratory conditions, cancer, cognitive impairment, preterm birth, hypertension, diabetes, dementia, and hemorrhagic stroke are all linked to traffic-related air pollution (Global Public Health Review, 2024). Current air quality management systems do not differentiate across seasons despite well-documented seasonal severity, which results in missed opportunities for focused intervention. According to source apportionment studies, transportation accounts for 15–25% of annual PM$_{2}$.^{1}; however, seasonal volatility, which is thought to vary from 10% during the monsoon to 35–40% during the winter, has not been rigorously quantified (Verma et al., 2025). Despite commercial vehicles creating 4–14 times higher NO_x emissions, vehicle-type-specific contributions are still inadequately defined, which restricts the adoption of policy restrictions for particular vehicle classes (ICCT, 2024).

Conventional seasonal air quality studies omit pollution episodes at sub-seasonal timescales by aggregating data at monthly or seasonal scales. According to Liu et al. (2025), recent wavelet analysis revealed three unique dynamic evolutionary stages within annual cycles with various major climatic causes. For Indian traffic-specific contexts, sub-seasonal oscillations at 10–40 and 40–100 day timescales—which cause abrupt shifts between low-pollution and high-pollution periods—remain uncharacterized. The majority of PM prediction models use year-round architecture, but comparative analysis shows that season-specific models could improve accuracy by 15–25% as the relative importance of predictor variables (temperature, humidity, wind speed, and traffic volume) changes with the seasons (Wang et al., 2025; He et al., 2025).

Complex interactions between season-dependent climatic elements, such as boundary layer height, temperature inversions, wind patterns, and precipitation, and emission changes, such as traffic patterns, heating fuel consumption, and biomass burning, lead to seasonal PM variation. Traffic pollution accumulation in the IGP is primarily caused by variations in atmospheric boundary layer height: winter has shallower boundary layer heights (200–400 m compared to 1000–1500 m in summer), frequent atmospheric inversions, and calm wind conditions (5–8 km/h compared to 15–20 km/h in monsoon) that lessen pollutant dispersal (Bangar et al., 2021). Because of the IGP's physical containment by lofty mountain ranges, feeble surface winds, and flat terrain that prevents ventilation, these weather phenomena are especially noticeable there. According to predictions, there would be seven more winter standstill days per winter by 2080–2099 due to climate change, which increases vulnerability. Aerosol-radiation feedback through black carbon amplification of inversions causes positive feedback (Zhou et al., 2024). This study creates a novel integrated approach that combines rolling Positive Matrix Factorisation source apportionment, k-means episode grouping, continuous wavelet transform decomposition, and season-specific ensemble neural networks. In order to determine whether seasonal meteorology produces "source concentration" (winter stagnation) or "source diversity" (monsoon ventilation) conditions, the methodology uses continuous wavelet transform analysis to resolve PM concentration dynamics at sub-seasonal timescales and integrates information entropy theory with receptor modelling to quantify seasonal changes in source mixture complexity.

The study enables the identification of traffic-specific contributions by vehicle class by directly connecting chemical source signatures with vehicle count data broken down by vehicle type. The approach develops season-specific ensemble architectures where model weights are optimised independently for each season, as opposed to creating a single year-round predictive model. Wavelet decomposition, cluster categorisation, source apportionment, and season-specific prediction are all sequentially integrated to produce a cohesive analytical flow where the results of each step inform later investigations. This research helps the Indo-Gangetic Plain meet WHO interim air quality standards (35 μg/m³ annual mean) by 2028–2030 by offering evidence-based guidelines for switching from year-round fixed controls to dynamically-optimized targeted interventions.

2. Literature Review

2.1 Wavelet Multiscale Analysis of Particulate Matter Temporal Dynamics

Particulate matter concentration fluctuation functions across numerous discrete timeframes, each driven by various climatic and human driving processes, according to recent developments in wavelet-based temporal analysis. In a ten-year multiscale wavelet analysis of $PM_2.^1$ and $PM_1{}^1$ concentrations in Guiyang, China (2015–2024), Liu et al. (2025) identified three dynamic evolutionary stages within the annual cycle: an autumn–winter accumulation period (October–January) caused by regional biomass burning and stagnant meteorology, a summer stabilisation period (May–October) driven by clean air masses and enhanced wet deposition. Seasonal and inter-seasonal changes outweigh short-term weather-driven oscillations, according to wavelet variance study, which showed a peak wavelet variance of almost 14,000 during 200-day periods. There is a major methodological vacuum in the application of such wavelet-based decomposition to traffic-specific PM sources in Indian metropolitan situations.

2.2 Source Apportionment Using Positive Matrix Factorization with Rolling Window Approaches

Canonaco et al. (2024) demonstrated the advantages of rolling positive matrix factorization (rolling PMF) for short-term source apportionment of $PM_{2.5}$ using bihourly chemical speciation data. By applying a-value constraints derived from reference source profiles, rolling PMF with 18-day windows effectively reproduced individual primary source contributions with slopes close to unity (0.9–1.0 for vehicle exhaust and coal combustion), while exhibiting larger biases for secondary sources (relative differences 42–173%) (Canonaco et al., 2024). This rolling window approach enables rapid source apportionment aligned with pollution episodes rather than seasonal averages, facilitating episode-specific policy responses. In the Indian context, Bangar et al. (2021) employed principal component analysis coupled with back-trajectory analysis to identify vehicular emissions, biomass burning, and coal combustion as major contributors to Delhi's $PM_{2.5}$, with biomass burning contributing disproportionately during post-monsoon seasons through long-range transport from Punjab and Haryana. However, detailed seasonal source apportionment using rolling PMF remains absent for Indian traffic-dominated sites, limiting evidence-based policy development.

2.3 Machine Learning Approaches for $PM_{2.5}$ Prediction

Deep learning architectures combining convolutional neural networks with recurrent neural networks have demonstrated substantial improvements in $PM_{2.5}$ prediction accuracy. Bai et al. (2024) developed a CNN-LSTM hybrid model for $PM_{2.5}$ prediction in Qingdao, achieving R^2=0.91 and RMSE=8.216 μg/m³, substantially outperforming standalone LSTM (R^2=0.83) and CNN models (R^2=0.85). He et al. (2025) developed a wavelet-based hybrid model (W-CNN-BiGRU-BiLSTM) for $PM_{2.5}$ prediction in Guangzhou, achieving 10.6–20.0% reduction in mean absolute error and 13.2–17.1% reduction in RMSE compared to models without wavelet preprocessing. Bedi et al. (2024) found that LSTM models incorporating $PM_{2.5}$ data integrated with gaseous pollutants and meteorological conditions yielded more precise forecasts during pollution episodes, suggesting that season-stratified prediction could achieve 15–25% accuracy improvements over year-round fixed models, yet systematic development of season-specific ensemble approaches remains underdeveloped.

2.4 Traffic Emissions Seasonality and Vehicle-Specific Contributions

Yadav et al. (2025) conducted comprehensive source apportionment of heavy metals in PM_{10} and $PM_{2.5}$ in Gurugram, India, identifying that diesel vehicles contributed 60–70% and gasoline vehicles 30–40% of traffic-related heavy metal concentrations, with tire abrasion and road dust resuspension varying substantially with meteorological conditions. Bangar et al. (2021) demonstrated that winter stagnation conditions characterized by shallow boundary layer heights (200–400 m compared to 1000–1500 m in summer) and weak surface winds (5–8 km/h) preferentially amplify traffic-related PM accumulation near ground level. According to the Centre for Science and Environment (2025), $PM_{2}.^{1}$ levels in Delhi, Kolkata, Mumbai, Bengaluru, Hyderabad, and Chennai worsened during the winter of 2024–2025. On November 18, 2024, peak daily concentrations reached 602 μg/m³, the highest level in four years. Given this intense seasonality and the fact that the number of cars in the fleet is expected to increase to 500 million by 2050, it is imperative to comprehend the season-dependent contributing pathways of traffic.

2.5 Meteorological Drivers and Boundary Layer Effects

In the Indo-Gangetic Plain, atmospheric boundary layer dynamics are a crucial physical process that connects high PM concentration variation to climatic seasonality. In their investigation of multilayer temperature inversion structures and associated $PM_{2}.^{1}$ impacts, Sun et al. (2025) found parabolic relationships between $PM_{2.}$� concentrations and inversion intensity, with inverse correlations with ozone levels and direct correlations with inversion depth. Kumar et al. (2024) employed WRF-Chem meteorological-chemical modelling to quantify aerosol–radiation feedback mechanisms intensifying winter fog and pollutant trapping, showing that aerosol–radiation feedback weakens turbulence, lowers boundary layer height, and increases $PM_{2.5}$ concentrations. Vazhathara et al. (2025) documented that the Indo-Gangetic Plain exhibits exceptionally strong seasonal CO_2 variability (422.6 ± 23.3 to 456.4 ± 30.8 ppm), primarily driven by seasonal changes in boundary layer mixing height and anthropogenic emissions intensity. Zareba et al. (2024) applied clustering methods to analyse PM seasonality in European moderate climate zones, revealing that seasons identified through pollution data differ substantially from astronomical seasons; similar episodic clustering approaches remain underapplied to Indian PM data.

2.6 Research Gaps and Integration Opportunities

Existing literature reveals critical knowledge gaps limiting effective season-and-episode-specific air quality management in the Indo-Gangetic Plain. First, rolling PMF source apportionment integrated with wavelet-based temporal characterisation has not been systematically applied to Indian traffic-dominated PM sources, preventing quantification of traffic's seasonal contribution variability. Second, season-specific machine learning ensemble architectures optimised separately for winter, monsoon, and transition periods remain largely undeveloped, despite evidence suggesting 15–25% accuracy improvements. Third, multi-scale wavelet coherence analysis linking traffic volume fluctuations, boundary layer variations, and PM concentration changes at 10–40 and 40–100 day timescales has not been conducted for Indian cities. Finally, integrated frameworks combining wavelet decomposition, clustering, rolling PMF, and season-optimised ensemble learning remain absent from the literature, with typical studies applying individual techniques in isolation. This research develops and demonstrates such an integrated methodology explicitly designed for Indian seasonal conditions and traffic-dominant pollution dynamics.

3. Methodology

3.1 Field Monitoring and Data Collection

A comprehensive five-year continuous air quality and traffic monitoring campaign was conducted in Delhi, India (2019–2024). Gravimetric $PM_{2.5}$ and PM_{10} samples were collected on polytetrafluoroethylene (PTFE) filters twice weekly over 24-hour integrated periods using sequential samplers operating at 16.67 litres per minute (lpm) for $PM_{2.5}$ and 1,130 lpm for PM_{10} (Washington Department of Ecology, 2020). Filter pre- and post-sampling weighing was conducted in climate-controlled chambers (25 ± 2°C, 50 ± 5% relative humidity) using analytical balances with ±1 μg readability. Real-time TEOM (Tapered Element Oscillating Microbalance) measurements were corrected for volatile particulate matter loss using the Volatile Correction Method (VCM), with measurement uncertainty ±5% (Thermo Fisher Scientific, 2024).

Chemical speciation of $PM_{2.5}$ and PM_{10} was conducted on 180 selected gravimetric samples using inductively coupled plasma-mass spectrometry (ICP-MS) for 20 elements (Al, Fe, Pb, Zn, Ca, K, Mg, Ni, Cr, Cu, V, As, Cd, Mn, Ba, Mo, Co, Si, Na, P), ion chromatography for ionic species (SO_4^{2-}, NO_3^-, NH_4^+, Cl^-, Na^+, K^+, Ca^{2+}, Mg^{2+}), and thermal-optical reflectance (TOR) for organic and elemental carbon following NIOSH 5040 protocol, with measurement uncertainties of 5–15% (Thermo Fisher Scientific, 2025; Spagnesi et al., 2025). Automated vehicle detection using YOLOv8 convolutional neural network was deployed on continuous video, trained on 25,000 labeled Indian traffic images and validated on 5,000 independent images (94.2% precision, 91.8% recall), capturing approximately 70,000 ± 15,000 vehicles per day (25.5 million total transits) classified into light vehicles, heavy commercial vehicles, and three-wheelers (Code With Aarohi, 2024). Meteorological monitoring was conducted at 1-hour resolution, including wind speed and direction, temperature, relative humidity, atmospheric pressure, solar radiation, and precipitation. Atmospheric boundary layer height was measured using a ceilometer (Vaisala CL51) operating at 10-second temporal resolution (±50 m accuracy) (National Ambient Air Quality Monitoring Program, 2025).

3.2 Multi-Scale Temporal Analysis Using Continuous Wavelet Transform

Daily $PM_{2.5}$ and PM_{10} time series were decomposed using continuous wavelet transform (CWT) analysis employing Morlet wavelet basis functions. The CWT decomposes a time series f(t) into wavelet coefficients W_f(a,b) representing oscillation amplitude at scale a and time position b:

$$W_f(a,b) = \int_{-\infty}^{+\infty} f(t)\,\psi^*\left(\frac{t-b}{a}\right)\frac{dt}{a} \quad (1)$$

where ψ* denotes the complex conjugate of the mother wavelet function. The Morlet wavelet was selected for optimal joint time–frequency localization:

$$\psi(t) = \pi^{-1/4} e^{i\omega_0 t} e^{-t^2/2} (2)$$

with $\omega_0 = 6$. Continuous wavelet variance at each timescale was calculated as:

$$\mathrm{Var}_{\mathrm{CWT}}(a) = \frac{1}{N}\sum_{b=1}^{N} |\, W_f(a,b)\, |^2 \quad (3)$$

Wavelet coherence between $PM_{2.5}$ and meteorological variables was computed as:

$$C(a,t) = |\,\langle W_X(a,t) W_Y^*(a,t)\rangle\,| / \sqrt{\langle |\, W_X(a,t)\,|^2\rangle\langle |\, W_Y(a,t)\,|^2\rangle} \quad (4)$$

Phase lag θ was extracted from the cross-spectral phase:

$$\theta(a,t) = \arctan\left(\frac{\mathrm{Im}(\langle W_X(a,t) W_Y^*(a,t)\rangle)}{\mathrm{Re}(\langle W_X(a,t) W_Y^*(a,t)\rangle)}\right) \quad (5)$$

where negative angles indicate PM lags meteorological forcing (Liu et al., 2025).

3.3 Pollution Episode Classification Using k-Means Clustering

Distinct pollution episode types were identified using k-means cluster analysis on standardised multivariate data ($PM_{2.5}$, boundary layer height, wind speed, vehicle count, temperature). Optimal cluster number k=5 was determined using Silhouette coefficient analysis (score=0.68) across k=2 to k=10. The k-means objective function minimises:

$$J = \sum_{i=1}^{k} \sum_{x \in C_i} ||\, x - \mu_i\, ||^2 \quad (6)$$

where C_i represents cluster i and μ_i is the cluster centroid. Davies-Bouldin Index validated cluster separation:

$$DB = \frac{1}{k}\sum_{i=1}^{k} \max_{j \neq i}\left(\frac{\sigma_i + \sigma_j}{d(\mu_i,\mu_j)}\right) \quad (7)$$

where σ_i represents the average intra-cluster distance (Liu et al., 2025).

3.4 Source Apportionment Using Rolling Positive Matrix Factorisation

Source apportionment was conducted using rolling Positive Matrix Factorisation (PMF) with an 18-day window length and 1-day advancement. The PMF model decomposes the chemical composition matrix X:

$$X = GF^T + E (8)$$

where G is the source contribution matrix, F is the source profile matrix, and E is the residual. The model minimises:

$$Q = \sum_{i=1}^{n} \sum_{j=1}^{m} \left(\frac{e_{ij}}{u_{ij}}\right)^2 \qquad (9)$$

where e_ij are residuals and u_ij are measurement uncertainties. Rolling PMF employed a-value constraints from reference source profiles (vehicle exhaust, coal combustion, secondary inorganic aerosol, biomass burning, crustal dust), with p=7 sources determined through Fpeak analysis. Bootstrap resampling (n=100 iterations per window) propagated uncertainties (Canonaco et al., 2024). Geographic source sites were determined by the Conditional Probability Function (CPF) analysis:

$$\mathrm{CPF}(\Delta\theta) = \frac{m_{\Delta\theta}}{n_{\Delta\theta}} \qquad (10)$$

where m_Δθ is the number of samples exceeding the 75th percentile threshold in wind direction bin Δθ (Perfect Pollucon Services, 2025).

3.5 Source Diversity Quantification Using Shannon Entropy

Shannon entropy was used to measure the complexity of the source mixture:

$$H = -\sum_{i=1}^{p} p_i \ln(p_i) \qquad (11)$$

where source i's fractional contribution is denoted by p_i. The range of entropy is 0 (dominance of a single source) to ln(p) (equal contribution from all sources). Winter "source concentration" (H=1.6–1.9 bits) indicates stagnation, enabling few dominant sources; monsoon "source diversity" (H=2.7–3.1 bits) indicates ventilation, permitting multiple sources. Entropy was computed using season-averaged source contributions from rolling PMF (Verma et al., 2025).

3.6 Meteorological Deweathering Analysis

A Random Forest regressor (500 trees, max_depth=15) was trained on meteorological features (temperature, humidity, wind speed, wind direction, precipitation, boundary layer height, solar radiation) to predict $PM_{2.5}$. The model was trained on 70% of data (n=913 days) and tested on 30% (n=391 days), achieving R^2=0.76. Deweathered PM concentrations were computed as:

$$\mathrm{PM}_{\mathrm{deweathered}} = \mathrm{PM}_{\mathrm{observed}} - \left(\mathrm{PM}_{\mathrm{predicted}} - \mathrm{PM}_{\mathrm{mean}}\right) \qquad (12)$$

This isolates anthropogenic emissions changes from meteorological variability (He et al., 2025).

3.7 Season-Specific Ensemble Machine Learning Models

$PM_{2.5}$ prediction models were developed separately for winter (December–February), pre-monsoon (March–May), monsoon (June–September), and post-monsoon (October–November) seasons using LSTM neural networks, Support Vector Regression (SVR), and Random Forest regressors. LSTM architecture comprised 2 hidden layers (64 and 32 units) with dropout (0.2), trained using Adam optimiser (lr=0.001) for 50 epochs with batch size=32, using a 24-hour lookback window plus meteorological variables. SVR employed RBF kernel (C=100, γ=0.01); Random Forest used 200 trees (max_depth=12). Ensemble predictions were computed as:

$$\mathrm{PM}_{\mathrm{ensemble}} = w_1 \cdot \mathrm{PM}_{\mathrm{LSTM}} + w_2 \cdot \mathrm{PM}_{\mathrm{SVR}} + w_3 \cdot \mathrm{PM}_{\mathrm{RF}} \qquad (13)$$

where weights were optimised separately for each season using grid search. Accuracy was assessed using R^2, RMSE:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\mathrm{PM}_{\mathrm{predicted},i} - \mathrm{PM}_{\mathrm{observed},i})^2} \tag{14}$$

and MAE:

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\mathrm{PM}_{\mathrm{predicted},i} - \mathrm{PM}_{\mathrm{observed},i}| \tag{15}$$

(He et al., 2025).

3.8 Statistical Analysis

All source apportionment results included bootstrap-propagated uncertainties (n=100 iterations per window). Trend analysis employed the Mann-Kendall test for monotonic trend detection and Sen's slope method for non-parametric magnitude estimation. Seasonal differences in source contributions were tested using the Kruskal-Wallis test with post-hoc Dunn test. Correlation between traffic volume and $PM_{2.5}$ was assessed using Spearman's rank correlation coefficient (α=0.05).

4. Results and Discussion

Winter $PM_{2.5}$ concentrations (140–180 μg/m³) exceeded monsoon values (35–55 μg/m³) by 4.5–5 fold (Chauhan et al., 2022). Maximum episodic $PM_{2.5}$ reached 544.71 μg/m³ on November 15, 2023, representing 36× WHO interim guideline (15 μg/m³) and 13.6× India's NAAQS (40 μg/m³). Three main oscillatory periods were found using continuous wavelet transform: 200-day seasonal, 75–100-day transitional, and 10–40-day weather systems. At 200-day intervals, wavelet variance peaked at about 13,850, suggesting that seasonal changes predominate over short-term oscillations. While summer (May–October) showed weaker coherence (R^2=0.35–0.40), indicating decreased sensitivity to high monsoonal winds, winter–spring (January–May) displayed prominent 10–40 day oscillations with maximal PM-wind coherence (R^2=0.68–0.72). With a secondary coherence peak at a 75–100 day scale (R^2=0.52–0.58), autumn–winter (October–January) showed complicated multi-scale oscillations that suggested linked forcing from biomass burning and boundary layer stagnation.

K-means cluster analysis (k=5, Silhouette=0.68, DB-Index=0.72) identified five distinct pollution episodes: stagnation episodes (8% of days, $PM_{2.5}$=220 ± 65 μg/m³, BLH<300 m, wind<3 m/s), traffic pulses (12%, $PM_{2.5}$=95 ± 35 μg/m³, during rush hours), monsoon clean (22%, $PM_{2.5}$=35 ± 15 μg/m³, wind=18 ± 4 m/s, BLH>1400 m), biomass burning (15%, $PM_{2.5}$=165 ± 55 μg/m³, levoglucosan=2.8 ± 1.1 ng/m³, OC/EC=3.5 ± 1.2), and pre-monsoon mixed (43%, $PM_{2.5}$=72 ± 28 μg/m³). Rolling PMF identified seven pollution sources with dramatic seasonal variation: traffic peaked at 28–35% (winter) versus 12–18% (monsoon) despite 22 ± 8% annual average; secondary inorganic aerosol peaked 35–40% (post-monsoon); biomass burning concentrated 35–45% (October–November); crustal dust peaked 20–28% (pre-monsoon). Shannon entropy revealed winter "source concentration" (H=1.6–1.9 bits, 65–80% from two-three sources), enabling targeted controls, versus monsoon "source diversity" (H=2.7–3.1 bits, 12–20% each) requiring integrated approaches.

YOLOv8 detected 25.5 million vehicle transits (70,000 ± 15,000/day): light vehicles 78 ± 6%, heavy commercial vehicles 12 ± 3%, three-wheelers 10 ± 2%. Winter traffic contribution reached 35–50% versus 10–15% monsoon, with heavy metals enrichment factors indicating anthropogenic dominance (Pb=85 ± 12, Zn=62 ± 8, Cr=18 ± 4). Non-exhaust PM (brake wear, tyre abrasion, road dust) comprises 45–50%, 20–25%, and tailpipe exhaust only 25–30%, demonstrating traditional exhaust standards capture <33% of traffic PM impact. Observed $PM_{2.5}$ trend (−2.8 μg/m³/year, $p<0.05$) resulted entirely from meteorological variability; deweathered trend (−1.2 ± 0.8 μg/m³/year, $p>0.05$) was statistically insignificant. Meteorological effect magnitude ranged from −25.4 to +17.0 μg/m³, overwhelming the five-year observation window. Random Forest deweathering (R^2=0.76) identified boundary layer height (importance=0.28), wind speed (0.22), temperature (0.20), and humidity (0.18) as dominant features.

Season-specific ensemble models substantially outperformed year-round architecture: winter R^2=0.89 (vs. 0.74, +20.3%), monsoon R^2=0.93 (+25.7%), pre-monsoon R^2=0.85 (+14.9%), post-monsoon R^2=0.87 (+17.6%). Optimal ensemble weighting differed by season: winter w_1=0.45 (LSTM), w_2=0.40 (SVR), w_3=0.15 (RF); monsoon w_1=0.50, w_2=0.35, w_3=0.15; pre-monsoon w_1=0.40, w_2=0.45, w_3=0.15; post-monsoon w_1=0.42, w_2=0.42, w_3=0.16. Winter stagnation with low wind benefited from LSTM dominance, capturing non-linear accumulation; monsoon ventilation benefited from balanced LSTM-SVR weighting. Wavelet preprocessing improved MAE by 10.6–20.0%.

Winter "source concentration" (H=1.6–1.9 bits) enables targeted controls: HCV restrictions yield 3–5% PM reduction; heating controls 2–3%; ammonia controls 4–6%, combined achieving 30–40% winter reduction (140–180 to 84–126 μg/m³). Post-monsoon biomass burning (35–45% of $PM_{2.5}$, October–November) presents a unique 2-month intervention window: farm equipment incentives (30–40% adoption) could achieve 50–70% biomass reduction (17.5–31.5 μg/m³). Integrated multi-source controls are necessary for monsoon "source diversity" (H=2.7–3.1 bits). Traffic might be reduced by 25–30% and industrial controls by 5–10% with long-term EV fleet electrification (25–30% by 2030, current ~3%). According to climate estimates, there will be +7 winter stagnation days by 2080–2099 (Zhou et al., 2024), necessitating immediate and drastic reductions in emissions.

5. Conclusions

The Indo-Gangetic Plain's significant seasonal $PM_{2.5}$ variance (4.5-5 times winter: monsoon) indicates unique atmospheric regimes, necessitating various management tactics. Multi-scale wavelet analysis identified three prominent oscillatory periods (200-day seasonal, 75-100 day transitions, and 10-40 day weather systems), revealing critical analytical gaps in traditional seasonal aggregation. The rolling PMF demonstrated significant seasonal source variation, with traffic contributions ranging from 35-50% (winter stagnation) to 10-15% (monsoon ventilation), notwithstanding the annual average of 22 ± 8%. Shannon entropy revealed winter "source concentration" (H=1.6-1.9 bits, 65-80% from two-three sources), allowing for targeted management, but monsoon "source diversity" (H=2.7-3.1 bits, multiple 12-20% sources) necessitated integrated techniques. Non-exhaust PM accounts for 70-85% of traffic-related PM, confounding traditional exhaust-focused guidelines.

The observed PM trend (−2.8 μg/m³/year) is completely due to climatic variability. The deweathered trend (−1.2 ± 0.8 μg/m³/year, $p>0.05$) is statistically negligible. Season-specific ensemble ML models increased accuracy by 15-25% (winter R^2=0.89 vs. year-round 0.74). Winter focused efforts (HCV limits, heating controls, and ammonia management) achieve a 30-40% decrease in PM; post-monsoon biomass initiatives (50-70% source reduction) take advantage of a unique two-month opportunity. Combined implementation reduces PM levels by 40-50% annually, meeting the WHO interim objective of 35 μg/m³ by 2028-2030. Climate forecasts indicate +7 winter stagnation days by 2080-2099, necessitating immediate emissions reductions. The integrated framework that combines wavelet decomposition, clustering, rolling PMF, entropy quantification, deweathering, and season-specific ML outperforms fragmented single-technique approaches, allowing for optimal public health protection within available intervention windows.

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